SUHAS PALLA

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EDUCATION

University of Illinois at Urbana-Champaign Urbana-Champaign, IL
B.S. in Industrial Engineering; Minor in Computer Science *Expected May 2029*
Honors: James Scholar **GPA:** 3.80 / 4.00 **Relevant Coursework:** Statics, Linear Algebra, Intro to CS II (C++),

TECHNICAL SKILLS

Design & Manufacturing: Fusion 360, SolidWorks, GD&T, Design for Manufacturability, Static FEA, Engineering Drawings, Tolerance & Fitment Analysis
Programming: Python, C++, Java, MATLAB, SQL, JavaScript/TypeScript
Software & Data: React, Node.js, Git, Docker, Firebase, Data Pipelines/ETL, Machine Learning, Pandas

ENGINEERING & DESIGN EXPERIENCE

Reverse-Engineered Drone Champaign, IL
CAD Design & Stress Analysis (Fusion 360) *Feb 2026 – May 2026*
- Reverse-engineered a commercial quadcopter with a 4-person team — disassembled the unit, measured components with digital calipers, and set reference datums to model 7 parts to scale in Fusion 360
- Integrated all 7 modeled parts into a single full-assembly model, resolving fitment, mating, and interference constraints across teammates' components to reach first-pass assembly with zero interference errors
- Ran static FEA on a drone leg under a 3 lbf load — defined constraints, meshed the part, and evaluated von Mises stress, displacement, and factor of safety across 2 candidate materials to drive material selection

Illini EV Concept Champaign, IL
Mechanical & Composites Team Member *Aug 2026 – Present*
- Designing drivetrain, suspension, steering, and carbon-fiber body components for a competition electric vehicle (EV3) in SolidWorks, applying FEA and material selection to reduce mass while meeting strength targets
- Manufacturing parts across the team machine shop (CNC mill, lathe, waterjet, 3D printing, welding, composite layup), applying design-for-manufacturability (DFM) to cut part count, machining time, and fabrication cost from CAD to assembly
- Producing GD&T-toleranced engineering drawings for in-house and vendor fabrication, tightening tolerance stack-ups to hit first-fit assembly and reduce rework across the mechanical subteam

ASME (American Society of Mechanical Engineers) Champaign, IL
Product Development Team Member *Feb 2026 – Present*
- Designing Version 2 of the REACH upper-extremity rehabilitation actuator (OnShape) — a tabletop stroke-recovery robot — to raise backdrivability under the off-axis patient loads that bound up the prior single-rail drive
- Prototyping and empirically benchmarking 3 linear rail-slider concepts on friction, cost, weight, and manufacturability, maximizing 3D-printed (PETG, CF-PLA) and off-the-shelf McMaster-Carr/MISUMI parts, and integrating an inline load cell and potentiometer to measure patient push/pull effort
- Engineering a tool-less quick-release handle latch and non-slip table-docking clamp that resists distal lift-off under load and reconfigures between left/right arm setups in under 2 minutes, then integrating the down-selected design into a showcase-ready unit with complete assembly work instructions

SOFTWARE EXPERIENCE

FiPet Remote
Head of App Development *May 2026 – Present*
- Lead development of a gamified financial-literacy app for teens (live on the App Store, 56+ five-star reviews), owning architecture across authentication, onboarding, and Firestore data workflows
- Shipped a production onboarding error-recovery system and authored 11+ backend integration tests (76/76 passing), eliminating silent failures across the account-creation flow
- Built a username-search Cloud Function (TypeScript/Firebase) enabling friend discovery via a normalized reservation model that returns minimal profiles and prevents private-field leakage

IDX Exchange Remote
Software Developer Intern *May 2026 – Aug 2026*
- Built and shipped a full-stack, Zillow-style property-search platform (React, Node.js/Express, MySQL) serving 53K+ live MLS listings through a filterable, paginated REST API
- Optimized backend queries with composite indexes (validated via EXPLAIN to eliminate full-table scans) and integrated the Anthropic Claude API to parse natural-language searches into structured filters
- Engineered the React frontend — image carousel/lightbox gallery, Google Maps, and client-side sort/filter — with Jest and React Testing Library suites reaching 70%+ coverage on critical paths

AI/ML Researcher Remote
Distracted Driving Detection *Jun 2024 – Jul 2024*
- Developed a computer-vision CNN to detect distracted-driving behaviors from in-vehicle image data, improving classification accuracy from 72% to 84% via architecture tuning and reproducible Python training pipelines

LEADERSHIP

ICON (International Consulting Network) Champaign, IL
Assistant Project Manager *Sep 2025 – Present*
- Evaluated AI-driven logistics solutions with an Automation / Augmentation / Speed-Gain / Error-Reduction framework, modeling projected speed gains and error reduction across transportation workflows and presenting recommendations to stakeholders